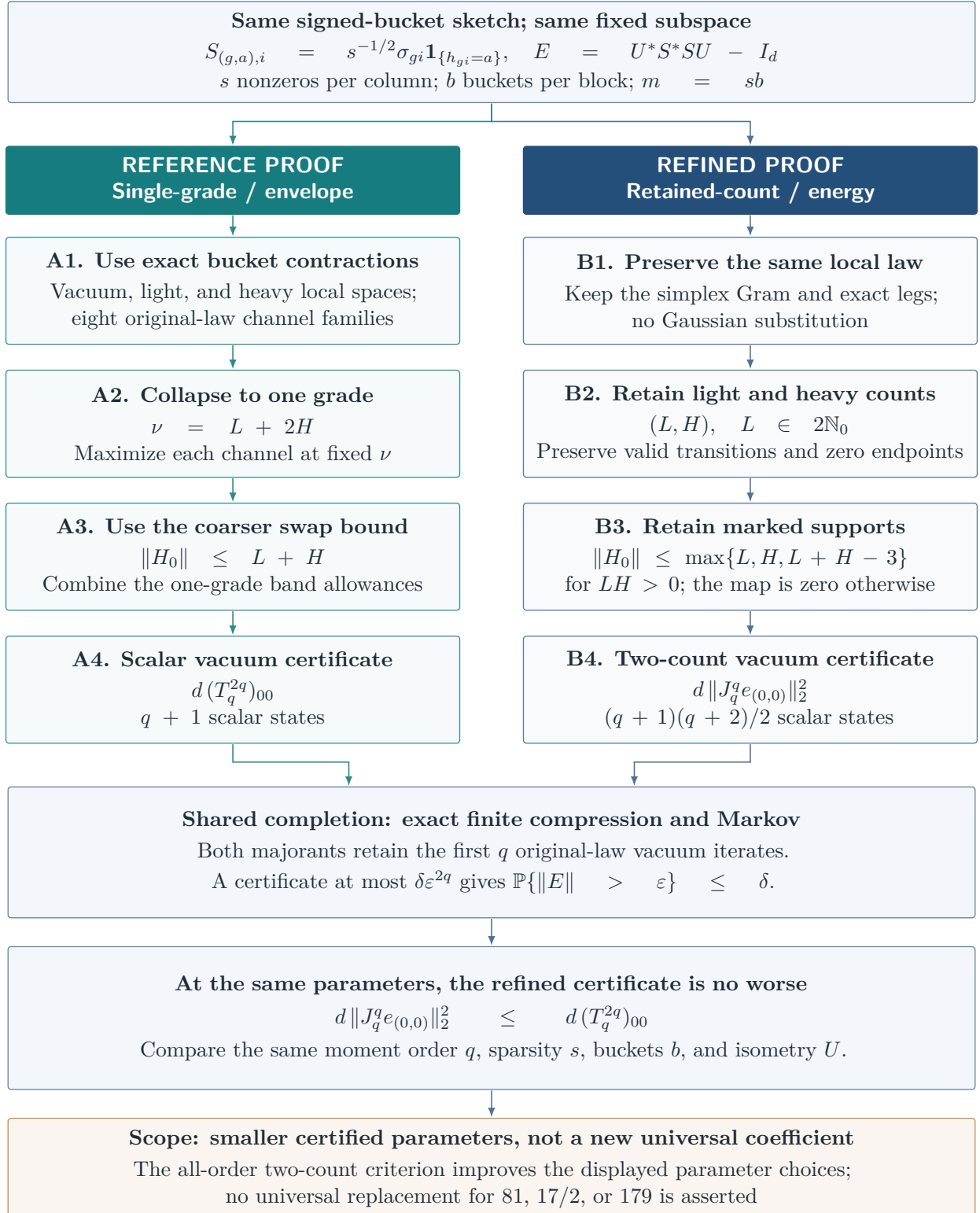


SparseStack

Where retaining both occupation counts sharpens the certificate

Read downward. Teal shows the reference proof; blue shows the refinement.

Full-width boxes identify the common model, completion, and scope.



Proof and constants comparison

Same signed-bucket law; retain the compatible count transitions.

The new argument uses the local channel bounds already in v11, including its sharper marked-support heavy-swap lemma. The added step keeps (L, H) through the vacuum walk instead of maximizing each channel separately at fixed ν .

Comparison	Reference proof Scalar-grade certificate	Refined proof Retained-count certificate
State retained	One grade $\nu = L + 2H$; $q + 1$ scalar states.	Both counts (L, H) ; $(q + 1)(q + 2)/2$ scalar states.
Local law	Exact signed-bucket multiplication.	Identical local law; no Gaussian moment substitution.
Heavy-swap bound	The direct uniform proof uses $\ H_0\ \leq L + H$.	Use $\max\{L, H, L + H - 3\}$ when $LH > 0$, and zero otherwise.
Propagation	A tridiagonal T_q uses separate channel maxima at each grade.	J_q retains the actual count endpoints, including forbidden transitions.
Moment certificate	$\mathbb{E} \operatorname{tr} E^{2q} \leq d(T_q^{2q})_{00}$.	$\mathbb{E} \operatorname{tr} E^{2q} \leq d\ J_q^q e_{(0,0)}\ _2^2$.
Comparison guarantee	Printed universal prescription follows from a geometric envelope.	At fixed d, q, s, b , the two-count certificate is no worse than the scalar one.
What is improved	Printed constants 81, 17/2, and 179 remain a valid fallback.	An all- q computable criterion and smaller certified examples; not new universal replacements.

Sufficient row counts at the same sparsity

$\varepsilon = 1/2$, $\delta = 0.01$; each row keeps the same q and s in every comparison. The old scalar certificate is evaluated directly to isolate the additional gain from retaining both counts.

d	q	s	Printed	Scalar T_q	Two-count J_q
10	5	187	6 919	748	561
100	7	255	37 485	5 865	4 335
1 000	9	323	330 429	58 786	43 928
10 000	10	357	3 246 915	669 732	503 370

Sufficient original-law certificates, not empirical thresholds. The proof, complete parameter comparisons, and arithmetic-check scope appear in Sections 3–5.

Quantitative endpoint and scope

$$d\|J_q^q e_{(0,0)}\|_2^2 \leq \delta \varepsilon^{2q} \implies \mathbb{P}\{\|E\| > \varepsilon\} \leq \delta.$$

At $d = 1000$, this lowers the row certificate from 58 786 to 43 928 with the same $s = 323$. A separate verified tradeoff uses $s = 152$ and $m = 51\,224$; it is not a universal new parameter prescription.

Result and scope

The direct SparseStack appendix already contains exact signed-bucket contractions and a sharper heavy-swap estimate. Its uniform constant proof discards part of this information by reducing two occupation counts to one grade. Retaining both counts gives an explicit, all-order vacuum certificate on a matrix of size $(q+1)(q+2)/2$.

For the same q and column sparsity, the certificate saves about 25% of the rows relative to direct numerical evaluation of the old scalar-grade certificate in the displayed examples. It also permits fewer nonzeros per column at still much smaller row counts than the printed prescription.

Scope: this note proves an all- q computable certificate and certified parameter improvements, not a smaller universal replacement for the printed numbers 81 and $17/2$. The local contractions and the marked-support lemma are credited to v11; the new contribution here is their retained-count assembly and its evaluated constants.

1 Model, baseline, and law preservation

Let $U : \mathbb{R}^d \rightarrow \mathbb{R}^n$ be a deterministic isometry with row vectors u_i^* , so $\sum_i u_i u_i^* = I_d$. Let $s \geq 1$, $b \geq 2$, and $m = sb$. In each block $g \in [s]$ and column $i \in [n]$, draw an independent uniform hash $h_{gi} \in [b]$ and an independent unbiased sign σ_{gi} . Define

$$S_{(g,a),i} = s^{-1/2} \sigma_{gi} \mathbf{1}_{\{h_{gi}=a\}}, \quad E = U^* S^* S U - I_d. \quad (1.1)$$

All columns have exactly s nonzeros and squared norm one. Both spectral edges are controlled by $\|E\|$. The analysis is uniform over the ambient dimension and every fixed U .

The reference is the *direct* law in [1, appendix “A direct SparseStack proof with explicit constants”], rather than its earlier independent-entry comparison. The public SparseStack preprint [2] is relevant background, but its reported Lean certificate does not certify the analytic refinements here.

Printed baseline.

For $0 < \varepsilon \leq 1$, $0 < \delta \leq 1$, v11 prescribes

$$q = \max\{1, \lceil \log_4(d/\delta) \rceil\}, \quad s_0 = \left\lceil \frac{(17/2)(2q+1)}{\varepsilon} \right\rceil, \quad b_0 = \left\lceil \frac{81(d+2q+1)}{s_0 \varepsilon^2} \right\rceil. \quad (1.2)$$

It proves failure probability at most δ , with $s_0 < 26.5q/\varepsilon$ and $m_0 = s_0 b_0 < 179(d+q)/\varepsilon^2$. These are the strongest direct uniform constants stated in the supplied version; we do not compare only against the much larger earlier reference constants.

Why ordinary Gaussian Wick substitution is invalid.

At one signed-bucket site define $e_a = \sqrt{b} \sigma \mathbf{1}_{\{h=a\}}$. For integers $t \geq 1$,

$$\mathbb{E} \prod_{j=1}^t e_{a_j} = \begin{cases} b^{t/2-1}, & t \text{ is even and all } a_j \text{ agree,} \\ 0, & \text{otherwise.} \end{cases} \quad (1.3)$$

In particular the fourth cumulant tensor is

$$\kappa_4(a, c, e, f) = b \mathbf{1}_{\{a=c=e=f\}} - \delta_{ac} \delta_{ef} - \delta_{ae} \delta_{cf} - \delta_{af} \delta_{ce}. \quad (1.4)$$

For a repeated collision edge, the fourth moment is $1/b$, not $3/b^2$; a four-cycle of collisions has weight $1/b^3$. These are already recorded in v11’s fourth-order component-reuse example.

Wick's Gaussian formula [3] is therefore only the pairing part of an original-law moment–cumulant expansion. Here we implement all local corrections exactly through the finite multiplication table. The transferable part of the rMPS hybrid [4] is *exact contractions followed by continuation-valid energy propagation*, not replacement of hashes or signs by Gaussians.

2 Exact local contractions and inherited channel estimates

Put $h = b - 1$ and $r_b = b/(b - 1)$. The orthogonal local levels are vacuum, light, and heavy:

$$\Omega = 1, \quad e_a = \sqrt{b} \sigma \mathbf{1}_{\{h_{gi}=a\}}, \quad f_a = \frac{b \mathbf{1}_{\{h_{gi}=a\}} - 1}{\sqrt{b-1}}.$$

The light vectors are orthonormal, whereas the heavy vectors form a simplex:

$$\langle f_a, f_c \rangle = \frac{b \delta_{ac} - 1}{b-1}, \quad \sum_a f_a = 0, \quad \sum_a |f_a\rangle \langle f_a| = r_b I. \quad (2.1)$$

Define local legs

$$P_a^* = |e_a\rangle \langle \Omega|, \quad P_a = |\Omega\rangle \langle e_a|, \quad R_a = |f_a\rangle \langle e_a|, \quad R_a^* = |e_a\rangle \langle f_a|.$$

Multiplication by e_a is exactly

$$J_a = P_a + P_a^* + \sqrt{h}(R_a + R_a^*). \quad (2.2)$$

Indeed $J_a \Omega = e_a$, $J_a e_c = \delta_{ac}(\Omega + \sqrt{h} f_a)$, and $J_a v = \sqrt{h} \langle f_a, v \rangle e_a$ on the heavy space. This is an original-law identity, not a moment approximation.

On the tensor product over sites (g, i) , let L_g, H_g be the light and heavy counts, and write

$$L = \sum_g L_g, \quad H = \sum_g H_g, \quad K = L + H, \quad \nu = L + 2H.$$

The multiplier of (1.1) is

$$\mathcal{M}_E = \frac{1}{m} \sum_{g,a} \sum_{i \neq j} u_i u_j^* \otimes J_{a,gi} J_{a,gj}. \quad (2.3)$$

Every term changes ν by $-2, 0$, or 2 . It also preserves even light parity in each sketch block. This invariant space contains the vacuum; in particular L is even.

2.1 The eight families

All sums below are over g, a and ordered $i \neq j$, with those subscripts suppressed:

$$\begin{aligned} L_+ &= \sum u_i u_j^* \otimes P_i^* P_j^*, & L_0 &= \sum u_i u_j^* \otimes (P_i^* P_j + P_i P_j^*), \\ X_+ &= \sum u_i u_j^* \otimes P_i^* R_j, & Y_+ &= \sum u_i u_j^* \otimes R_i P_j^*, \\ X_0 &= \sum u_i u_j^* \otimes P_i^* R_j^*, & Y_0 &= \sum u_i u_j^* \otimes R_i^* P_j^*, \\ H_+ &= \sum u_i u_j^* \otimes R_i R_j, & H_0 &= \sum u_i u_j^* \otimes (R_i R_j^* + R_i^* R_j). \end{aligned} \quad (2.4)$$

Thus

$$\begin{aligned} mB^+ &= L_+ + \sqrt{h}(X_+ + Y_+) + hH_+, \\ mB^0 &= L_0 + \sqrt{h}(X_0 + X_0^* + Y_0 + Y_0^*) + hH_0, & B^- &= (B^+)^*. \end{aligned} \quad (2.5)$$

Lemma 2.1 (Inherited original-law estimates). *On an even-block sector of total counts (L, H) , the following bounds hold. A family is zero at every endpoint listed as forbidden.*

Family	Output counts	Norm bound	Condition
L_+	$(L + 2, H)$	$\sqrt{(m + L)(d + L + 1)}$	<i>always</i>
X_+	$(L, H + 1)$	$\sqrt{(H + L/2)(d + L - 1)}$	$L \geq 2$
Y_+	$(L, H + 1)$	L	$L \geq 2$
H_+	$(L - 2, H + 2)$	$\sqrt{r_b(K - 1)(H + L/2)}$	$L \geq 2$
X_0	$(L + 2, H - 1)$	$\sqrt{r_b K(d + L + 1)}$	$H \geq 1$
Y_0	$(L + 2, H - 1)$	$\sqrt{r_b H(L + 2)}$	$H \geq 1$
L_0	(L, H)	$d + 2L - 1$	$L \geq 2$
H_0	(L, H)	$C(L, H) := \max\{L, H, L + H - 3\}$	$LH > 0$

Proof and attribution. The first seven bounds are the family table in [1, direct SparseStack appendix, “Collected Grams and parity” and “All channel bounds”]. We recall the operator inputs so that the imported estimate is precisely specified.

For $S_{ga}(Z) = \sum_i u_i \otimes Z_{a,gi}$, let $G(Z) = \sum_{ga} S_{ga}(Z) S_{ga}(Z)^*$. The shared-factor light Gram gives $G(P^*) \preceq (d + L - 1)I$ at $L > 0$, while $G(P) \preceq (m + L)I$. The heavy Grams satisfy

$$G(R^*) \preceq (H + L/2)I, \quad G(R) \preceq r_b K I.$$

On the target of H_+ , two new heavy sites lie in the same block, improving the latter bound to $r_b(K - 1)I$ with the source value of K . Block row–column Cauchy–Schwarz applied to the leg factorizations gives the L_+, X_+, H_+, X_0 bounds. The coordinate-frame hop bounds give Y_+, Y_0 , and $L_0 = G(P^*) - D_1 + H'$ gives its listed allowance. These are full operator bounds on the specified invariant sectors, so taking adjoints is legitimate.

For H_0 we use the stronger result already proved in [1, appendix “Exact bucket-sector calculations”, marked-support heavy-swap proposition], rather than the weaker K used in its uniform proof. Here is the relevant argument. In the coordinate frame $U = I_n$, the two orientations of a heavy swap have orthogonal marked-external supports: the marked index is heavy for one orientation and light for the other. Their norms in block g are at most L_g and H_g , respectively, so their sum has norm at most $\max(L_g, H_g)$, not $L_g + H_g$.

The identity

$$F(U) = (U^* \otimes I) F(I_n) (U \otimes I)$$

preserves the bound after compression. Summing contributing blocks gives $\sum_{g: L_g H_g > 0} \max(L_g, H_g)$. If all such blocks are light-dominated this is at most L ; if all are heavy-dominated it is at most H . Otherwise the minima subtracted from total occupancy include at least one from a light-dominated block and at least two from a heavy-dominated block, because positive L_g is even. The result is at most $L + H - 3$. This proves the final bound and its zero endpoints. $\square$

Remark 2.2 (What is and is not new). No improvement to the individual shared-factor or heavy-pair lemma is asserted. The finite bucket contractions and the sharper heavy-swap bound above were present in the supplied manuscript. The next step retains their compatible count labels throughout the vacuum evolution instead of maximizing each contribution independently at fixed ν .

3 A two-count certificate with explicit entries

For $q \geq 1$, define

$$\mathcal{I}_q = \{(L, H) : L \in 2\mathbb{N}_0, H \in \mathbb{N}_0, L + 2H \leq 2q\}. \quad (3.1)$$

Its size is $(q+1)(q+2)/2$. Physical configurations excluded by the ambient dimension may be left in this scalar index set: doing so only enlarges the nonnegative majorant.

Let $J = J_q(d, s, b)$ be a symmetric nonnegative matrix on $\mathcal{I}_q$. Set $m = sb$, $h = b - 1$, and $K = L + H$. Its diagonal is

$$J_{(L,H),(L,H)} = \frac{(d + 2L - 1)\mathbf{1}_{\{L>0\}} + hC(L, H)\mathbf{1}_{\{LH>0\}}}{m}. \quad (3.2)$$

Its undirected edges are the following, together with their symmetric counterparts:

$$J_{(L+2,H),(L,H)} = \frac{\sqrt{(m+L)(d+L+1)}}{m}, \quad (3.3)$$

$$J_{(L,H+1),(L,H)} = \frac{\sqrt{h(H+L/2)(d+L-1)} + L\sqrt{h}}{m}, \quad L \geq 2, \quad (3.4)$$

$$J_{(L-2,H+2),(L,H)} = \frac{\sqrt{bh(K-1)(H+L/2)}}{m}, \quad L \geq 2, \quad (3.5)$$

$$J_{(L+2,H-1),(L,H)} = \frac{\sqrt{bK(d+L+1)} + \sqrt{bH(L+2)}}{m}, \quad H \geq 1. \quad (3.6)$$

An entry is inserted only when both endpoints belong to $\mathcal{I}_q$; all other entries are zero. Each listed orientation determines a different undirected edge, so no edge is counted twice. In particular $J_{00} = 0$ and the only vacuum edge is

$$J_{(2,0),(0,0)} = \sqrt{(d+1)/m}.$$

Theorem 3.1 (Retained-count moment and probability certificate). *For every $d, q, s \geq 1$, $b \geq 2$, every ambient dimension $n \geq d$, and every fixed real isometry U , the sketch in (1.1) satisfies*

$$\boxed{\mathbb{E} \operatorname{tr} E^{2q} \leq d \|J_q(d, s, b)^q e_{(0,0)}\|_2^2}. \quad (3.7)$$

Consequently the fully explicit condition

$$\boxed{d \|J_q(d, s, b)^q e_{(0,0)}\|_2^2 \leq \delta \varepsilon^{2q}} \quad (3.8)$$

is sufficient for $\mathbb{P}\{\|E\| > \varepsilon\} \leq \delta$.

Proof. Work on the original product space and let Π retain block-even light parity and total grade $\nu \leq 2q$. Starting at the vacuum, one multiplication changes ν by at most two, so

$$(\Pi \mathcal{M}_E \Pi)^j(e_a \otimes \Omega) = \mathcal{M}_E^j(e_a \otimes \Omega), \quad 0 \leq j \leq q.$$

The exact multiplication identity from [1] is therefore

$$\mathbb{E} \operatorname{tr} E^{2q} = \sum_{a=1}^d \|(\Pi \mathcal{M}_E \Pi)^q(e_a \otimes \Omega)\|^2.$$

Decompose this cutoff by (L, H) . Lemma 2.1 and the multipliers in (2.5) give exactly (3.2)–(3.6). For example the two mixed-creation families have the same target, so their allowances add inside (3.4); the two heavy legs contribute $h\sqrt{r_b} = \sqrt{bh}$, as in (3.5). The within-grade pair contributes $\sqrt{h}\sqrt{r_b} = \sqrt{b}$, as in (3.6).

For a vector f , set $v(f)_{L,H} = \|\Pi_{L,H}f\|$. Orthogonality of the count sectors and the triangle inequality at each actual target imply

$$v((\Pi\mathcal{M}_E\Pi)f) \leq Jv(f)$$

componentwise. Induction gives $v(M^q(e_a \otimes \Omega)) \leq J^q e_{(0,0)}$. Square, sum the sectors and the d external seeds, and obtain (3.7). Finally $\|E\|^{2q} \leq \text{tr } E^{2q}$ for a self-adjoint matrix, so Markov gives (3.8). $\square$

3.1 Why this continuation is valid

Only orthogonal occupation counts are retained in the scalar majorant. They are not required to be invariant one at a time: the matrix J explicitly records their transitions. The full block-even space is invariant, and all local coefficients are computed on the original law. No discarded spectrum is assumed to survive a reshaping, and no scalar surrogate is iterated as if it were an exact state. This is a compatible energy majorant, not a quotient claiming to reconstruct the full random matrix.

4 Comparison with the old scalar-grade certificate

The direct appendix collapses the counts to $\nu = 2k$. In its notation the scalar matrix T_q has vacuum edge $\alpha_0 = \sqrt{(d+1)/m}$, vacuum diagonal zero, and

$$\begin{aligned} m\alpha_\nu &= \sqrt{(m+\nu)(d+\nu+1)} + \sqrt{h\nu(d+\nu-1)/2} + \nu\sqrt{h} \\ &\quad + \sqrt{bh\nu(\nu-1)/2}, \quad \nu > 0, \end{aligned} \quad (4.1)$$

$$m\xi_\nu = d + 2\nu - 1 + 2\sqrt{b(\nu-1)(d+\nu-1)} + (\nu+1)\sqrt{b} + h\nu. \quad (4.2)$$

Its diagonal is ξ_{2k} and its edge $k \leftrightarrow k+1$ is α_{2k} . These original local estimates are valid for every $b \geq 2$; $b \geq 10$ is only part of the later global parameter prescription.

Proposition 4.1 (No worse than the existing scalar certificate). *For the same d, q, s, b ,*

$$\|J_q^q e_{(0,0)}\|_2^2 \leq (T_q^{2q})_{00}. \quad (4.3)$$

Thus the original printed parameter prescription remains available as a universal fallback. The new certificate retains its asymptotic sufficiency, and can certify strictly smaller finite parameters.

Proof. Group the scalar coordinates of J_q by ν . Each family in (3.3)–(3.5) maps distinct source count pairs to distinct targets. Its norm between two grades is at most its largest weight. Triangle inequality for these families gives (4.1); here $L \leq \nu$, $H + L/2 = \nu/2$, and $K - 1 \leq \nu - 1$.

Within a grade, the diagonal and the two orientations of (3.6) are bounded by (4.2), exactly as in the appendix. The sharper $C(L, H)$ is no greater than K . For the Y_0 term use $H(L+2) \leq (\nu+1)^2/4$, while the X_0 endpoint $H \geq 1$ gives $K \leq \nu - 1$ and $d + L + 1 \leq d + \nu - 1$. Applying the same block-norm induction to this grouped scalar matrix gives (4.3). $\square$

4.1 Same sparsity: isolate the retained-count gain

Table 1 uses $\varepsilon = 1/2$, $\delta = 0.01$, and the same q, s_0 in all columns. The intermediate column evaluates the old T_q certificate rather than the paper’s geometric-envelope prescription. This separates the gain from numerical evaluation alone from the additional gain from retaining counts.

d	q	s_0	Printed m_0	Old T_q evaluated	Retained-count J_q
10	5	187	6 919	748	561
100	7	255	37 485	5 865	4 335
1 000	9	323	330 429	58 786	43 928
10 000	10	357	3 246 915	669 732	503 370

Table 1: Sufficient row counts under the original signed-bucket model, uniform over every fixed isometry of the displayed dimension. Only the last comparison isolates the retained-count refinement.

The retained-count savings relative to the evaluated old certificate are 25.0%, 26.1%, 25.3%, and 24.8%, respectively. At $d = 1000$, the improvement is $58\,786 \rightarrow 43\,928$ rows at exactly the same $s = 323$.

To separate two sources of this gain, using the two-count matrix but restoring the weaker $\|H_0\| \leq L + H$ gives 561, 4 335, 44 251, and 505 869 rows. Thus most of the gain is from retaining the count transitions and their zero endpoints. The already-known marked-support bound supplies an additional, smaller improvement; it is not presented as a newly discovered contraction.

4.2 A sparsity–dimension tradeoff

The certificate also permits smaller s , rather than merely fewer rows at fixed s . The following are verified choices, not a new all-parameter prescription:

d	Old s_0	Old printed m_0	New s	New b	New $m = sb$
10	187	6 919	88	7	616
100	255	37 485	120	40	4 800
1 000	323	330 429	152	337	51 224
10 000	357	3 246 915	168	3 555	597 240

Table 2: The same ε, δ, q as Table 1. Every new row is checked directly in (3.8). No extrapolation in d or q is used.

The values of s in this table equal $\lceil 4(2q + 1)/\varepsilon \rceil$ for these four inputs. That numerical pattern is *not* asserted sufficient with one universally reduced row coefficient.

5 Certificate verification and computational scope

The accompanying script `hybrid_appendices_checks.py` constructs every entry from rational numbers and square roots of rational numbers. A square root $\sqrt{a/b}$ is enclosed on a decimal grid of step 10^{-44} using integer square roots; the inequalities for the lower and upper endpoints are verified by integer squaring. Positive matrix-vector recurrence then encloses $d\|J^q e_0\|^2/\varepsilon^{2q}$ using rational arithmetic throughout.

For example the upper failure allowances at the four retained-count choices of Table 1 are, rounded for display,

$$0.004506489, \quad 0.008313043, \quad 0.009906864, \quad 0.009993962,$$

all below 0.01. The script records the full enclosures. These are deterministic theorem certificates, not Monte Carlo estimates.

5.1 Checking the smaller integer choices

At a fixed s , the certificate is not assumed monotone in b near $b = 2$: some heavy-channel weights increase with b . The following observation justifies the finite threshold checks.

Lemma 5.1 (Eventual monotonicity of the positive certificate). *At fixed d, q, s , both $(J_q^{2q})_{00}$ and $(T_q^{2q})_{00}$ are nonincreasing for real $b \geq \max\{2, 2q - 1\}$.*

Proof. The vacuum edge is proportional to $b^{-1/2}$. Every other positive summand of an edge or diagonal in the matrices above has logarithmic derivative at most $1/[b(b-1)]$ for $b \geq 2$. Indeed the potentially increasing factors are $(b-1)/b$ and $\sqrt{(b-1)/b}$; the remaining factors decrease, including $\sqrt{sb + L}/b$ and $\sqrt{b-1}/b$.

Expand a returning walk weight into positive summands. A walk of length $2q$ has at least two vacuum edges, so each product has logarithmic derivative at most

$$-\frac{1}{b} + \frac{2q-2}{b(b-1)} = \frac{2q-1-b}{b(b-1)} \leq 0.$$

Positive summation proves the claim. Zero terms may simply be omitted. $\square$

The script checks every smaller integer $2 \leq b < 2q - 1$ and checks the preceding integer at the reported threshold with a rational lower enclosure. Together with the lemma, this certifies the smallest integer $b \geq 2$ satisfying each displayed matrix criterion at the selected s . This is minimality for the *specified sufficient criterion*, not for the actual embedding problem.

5.2 Additional original-law checks

Independent finite enumeration is included for a rational 3×2 isometry, one sketch block, $b = 2, 3$, and moment orders 2, 4, 6. It evaluates the true hash/sign matrix moments without using the occupation recurrence, and checks them against the new majorant. The local fourth cumulant is also checked for every bucket-label pattern at $b = 2, \dots, 7$.

The matrix has $O(q^2)$ states and bounded degree, so one certificate takes $O(q^3)$ arithmetic operations with a sparse implementation of q matrix-vector steps. Bit complexity additionally depends on the enclosure precision and on q . No enumeration of the ambient dimension or the full occupation space is required for the uniform criterion.

6 What the hybrid improves, and what remains unchanged

The retained object is now the pair (L, H) , together with the original-law transition costs and the marked-support heavy-swap maximum. The update is an explicit positive matrix, and the moment remains a vacuum energy. This is the same exact-contraction/valid-continuation principle used in the rMPS hybrid, instantiated on a different local law.

The following limits are essential. The local estimates are not newly proved optimal for arbitrary U ; no improved worst-case sharpness result is asserted. The smaller examples do not by themselves establish universal replacements for 81, 17/2, or 179. The all- q theorem is (3.7), with the original prescription retained through (4.3). A further all-parameter constant theorem would need an analytic bound on this two-count vacuum expression, not just a finite numerical scan.

The proofs and arithmetic certificates here are analytic work. They are not an external review or a Lean replay. In particular the formal certificate associated with the earlier SparseStack theorem must not be described as certifying this note.

References

- [1] Diar Heidary. *Graded Multiplication Operators for Random-Matrix Moments*. Version 11, exposition and attribution revision, 16 September 2026. Supplied source `v11_attribution_revised.tex`. Used here: exact finite compression; nonnegative vacuum majorants; direct SparseStack family bounds; marked-support heavy swaps. These source lemmas are the inherited proof inputs.
- [2] Diar Heidary. *SparseStack Is an Optimal Oblivious Subspace Embedding*. arXiv:2609.02978v2, 9 September 2026. <https://arxiv.org/abs/2609.02978v2>. Background on the earlier original theorem and its reported formalization; its constants are not the baseline used in Table 1.
- [3] Leon Isserlis. On a formula for the product-moment coefficient of any order of a normal frequency distribution in any number of variables. *Biometrika* 12(1–2), 134–139, 1918. <https://doi.org/10.1093/biomet/12.1-2.134>. Gaussian pairing identity only, not a theorem for the bucket law.
- [4] Method Problems working note. *Gaussian rMPS Moments: Two Proof Chains and a Wick–Energy Hybrid*. 16 September 2026. Supplied source `rmeps_three_chains_hybrid.tex`. Motivation for retaining coefficient information through a proved boundary update; no Gaussian-core lemma is transferred without changing its law.